BRACKETED

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TEAM MEMBERS

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BACKGROUND

- Sports and esports fans often participate in pick'em or bracket-style challenges, but existing platforms are typically limited to a single league or event.
- Bracketed is a centralized website for sports and Esports pick'em games. Our goal is to create a platform that allows users to make predictions, track progress, and engage with multiple leagues in one location while competing with friends in leaderboards.

GOALS

- Combine traditional sports and Esports into a single competitive pick'em interface
- Create a clean, intuitive UI focused on engagement and usability
- Implement leaderboards and user accounts to foster community competition
- Build a scalable backend with database integration for real-time results and scoring

INTELLECTUAL MERITS

- League-agnostic backend architecture allowing scalable expansion
- Standardized API integration across different sports and esports
- Modular system design separating frontend, backend, and data layers
- Database schema supporting future league additions
- Website hosted for easy access to all users

BROADER IMPACTS

- Encourages community engagement around sports and esports
- Creates friendly competition among users
- Supports inclusivity by integrating both traditional sports and esports
- Demonstrates scalable software architecture practices
- Can be extended for educational, charity, or campus competitions

SYSTEM OVERVIEW

- Frontend (Client)
  - User interface
  - Challenge management
  - Prediction submission
  - Settings & navigation

- Backend (Server/API Layer)
  - League data retrieval
  - Authentication
  - Challenge management

- Database
  - Users
  - Challenges
  - Predictions
  - Game/match data

TECHNOLOGIES

- Frontend
  - React
  - JavaScript
- Backend
  - Node.js
  - League API integrations
- Database
  - PostgreSQL database hosted on Neon
- Other Tools
  - Git/GitHub for version control
  - Localhost development environment

RESULTS

- Completed
  - Working backend connected to frontend
  - Generalized League API integration
  - Frontend prediction submission
  - Connecting backend and database
  - Challenge creation
- Remaining
  - User authentication
  - Challenge invitation
  - Third-party API automation

CONSTRAINTS

Economic

- Free or low-cost APIs minimize expenses
- Minor power costs for hosting and testing
- Low cost for website hosting

Legal

- Restricted use of league logos / names
- Only use publicly available data via approved APIs
- Future expansion would require licensing or data agreements

Security

- Login system introduces data privacy risks
- Store minimal user info
- Encourage strong passwords
- Apply standard database protections

Ethical

- Few immediate concerns
- Must monitor for harassment or toxic behavior
- Avoid promoting gambling or exploitative practices

CHALLENGES

- We needed an efficient way to show the teams without using their logos due to legal issues
  - Solution: Simplified display to only show names/colors
- External API Limitations
  - Free tiers limit how many requests can be made
  - Solution: Save API data to our own database to reduce requests
- Generalizing League Data
  - Different APIs return different data formats
  - Solution: Standardized backend schema